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Batch 77

1. Function: printf()

Header: stdio.h

Algorithm

1. Start
2. Use printf() to display a message
3. End

Pseudocode

START

PRINTF “Hello World”

END

```
1 #include<stdio.h>
2 int main(){
3     printf("Hello World");
4     return 0;
5 }
```

Result

```
Hello World
-----
Process exited after 0.1778 seconds with return value 0
Press any key to continue . . .
```

2. Function: scanf()

Header: stdio.h

Algorithm

1. Start

2. Ask user for a number
3. Read number using scanf()
4. Print number
5. End

Pseudocode

START

INPUT num

PRINT num

END

```
1 #include<stdio.h>
2 int main() ){
3 int num;
4 printf("Enter a number:");
5 scanf("%d", &num);
6 printf("You entered: %d", num);
7 return 0;
8 }
```

```
Enter a number:4
You entered: 4
-----
Process exited after 2.716 seconds with return value 0
Press any key to continue . . .
```

3. Function: toupper()

Header:ctype.h

Algorithm

1. Start
2. Input a character
3. Convert to uppercase using toupper()

4. Print result

5. End

```
1 #include <stdio.h>
2 #include <ctype.h>
3 int main() {
4     char ch;
5     printf("Enter a letter: ");
6     scanf(" %c" ,&ch);
7     printf("Uppercase: %c", toupper(ch));
8     return 0;
9 }
```

```
Enter a letter: s
Uppercase: S
-----
Process exited after 2.648 seconds with return value 0
Press any key to continue . . .
```

4. Function: isdigit()

Header: ctype.h

```

1  #include<stdio.h>
2  #include<ctype.h>
3  int main() {
4      char ch;
5      printf("Enter a character: ");
6      scanf("%c", &ch);
7
8      if(isdigit(ch))
9          printf("Digit");
10     else
11         printf("Not a digit");
12
13     return 0;
14 }

```

```

Enter a character: s
Not a digit
-----
Process exited after 2.131 seconds with return value 0
Press any key to continue . . .

```

5. Function: strlen()

Header:string.h

```
1 #include<stdio.h>
2 #include<string.h>
3
4 int main() {
5     char str[50];
6     printf("Enter a word:");
7     scanf("%s", str);
8     printf("Length = %lu", strlen(str));
9     return 0;
10 }
```

```
Enter a word:raj
Length = 3
-----
Process exited after 2.755 seconds with return value 0
Press any key to continue . . .
```

6.Function strcpy()

Header:string.h

```
1  #include<stdio.h>
2  #include<string.h>
3
4  int main() {
5      char a[50], b[50];
6      printf("Enter a string");
7      scanf("%s", a);
8      strcpy(b, a);
9      printf("Copied string: %s", b);
10     return 0;
11 }
```

```
Enter a string sankalp
Copied string: sankalp
-----
Process exited after 6.694 seconds with return value 0
Press any key to continue . . .
```

7.Function: sqrt()

Header:math.h

```
1  #include<stdio.h>
2  #include <math.h>
3  int main(){
4      double n;
5      printf("Enter number: ");
6      scanf("%lf", &n);
7      printf("squre root = %21f", sqrt(n));
8      return 0;
9
10 }
```

```
Enter number: 5
squre root =          2.236068
-----
Process exited after 2.953 seconds with return value 0
Press any key to continue . . .
```

8.Function: malloc()

Header:stdlib.h

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 int main(){
4     int *p = (int *)malloc(sizeof(int));
5     *p = 20;
6     printf("Value = %d",*p);
7     free(p);
8     return 0;
9 }
```

Value = 20

Process exited after 0.1199 seconds with return value 0

Press any key to continue . . .

9. Function: free()

Header:stdlib.h

(Used together with malloc)

Explanation:

Free (ptr) releases memory that was allocated using malloc()

10.Function:assert()

Header:assert.h

```
1  #include <stdio.h>
2  #include <assert.h>
3  int main(){
4      int age = 18;
5      assert(age >= 18);
6      printf("Valid age\n");
7      return 0;
8  }
```

```
Valid age
```

```
-----
```

```
Process exited after 0.181 seconds with return value 0
```

```
Press any key to continue . . . █
```

Q1. Static Library vs Shared Library

Static Library (.a / .lib)

- Added to the program during compilation
- Final executable becomes bigger
- Faster execution
- No need for library file during runtime

Shared Library (.so / .dll)

- Linked during program execution
- Reduces executable size
- Same library can be used by many programs
- Requires library file during runtime

Q2. Dynamic Memory vs Static Memory

Static Memory

- Allocated at compile time
- Fixed size
- Example:

```
int a[10];
```

Dynamic Memory

- Allocated at runtime
- Flexible size
- Uses malloc(), calloc(), realloc()
- Must use free() to release memory

Example:

```
int *p = malloc(5 * sizeof(int));
```

Q3. Garbage Collection

- C does not have automatic garbage collection
- Programmer must manually free memory
- Use free() for memory allocated by malloc()

- Prevents memory leaks